

16.1: Working with Probability

Experiment - A process which yields an outcome.
Event - A set of outcomes
Sample Space - Set of all possible Outcomes.
Complement: Set of outcomes not contained in an event.
Mutually Exclusive: Two Events where there is no common outcome
Independent: The Occurence of one event does not affect another.

$P(A) = \frac{n(A)}{n(T)}$, value between 0 (impossible) & 1 (certain)

Can be estimated theoretically or experimentally.

Laws of Probability:

$P(A') = 1 - P(A)$
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (not mutually exclusive)
$P(A \cup B) = P(A) + P(B)$ (mutually exclusive)
$P(A \cap B) = P(A) \times P(B)$

Expectation:
$E(X) = nP(X)$, for a binary event.
Using Selections:
For a none replacement, choosing cycle out of N options, n draws, k success options, & k real successes:

$X \sim Hyp(N, k, n)$
$\Rightarrow P(X = k) = \frac{{}^{k}C_{k} \times {}^{N-k}C_{n-k}}{{}^{N}C_{n}}$

Aka Hypergeometric Distribution

16.2: Probability Distributions

A probability distribution is made when potential outcomes for an event (& their probabilities) are recorded
The event occuring is represented as a Random Variable, X, and the possible values X can take

x	u_1	u_2	.....	u_{n-1}	u_n
$P(x=x)$	$P(u_1)$	$P(u_2)$		$P(u_{n-1})$	$P(u_n)$

$E(X) = \sum_{1}^{n} x P(X = x)$

Risk:
In real life, probability can be described as risk, especially when it is an event which we do not want to happen.